



Q1 Student Database

5 marks

Write a Python program to manage a student database using sets.

Sample Input/Output

Add Student

Input

Enter your choice: 1

Enter student name: Ravi

Output

Student added successfully.

Add Another Student

Input

Enter your choice: 1

Enter student name: Priya

Output

Student added successfully.

Display Students

Input

Enter your choice: 4

Output

Students: {'Ravi', 'Priya'}

Search Student



Your Code

```
1 def main():
2     students = set()
3
4     while True:
5         print("\nStudent Database using Set")
6         print("1. Add Student")
7         print("2. Remove Student")
8         print("3. Search Student")
9         print("4. Display Students")
10        print("5. Count Students")
11        print("6. Exit")
12
13        try:
14            choice = int(input("Enter your choice: "))
15        except ValueError:
16            print("Invalid input. Please enter a number.")
17            continue
18
19        if choice == 1:
20            name = input("Enter student name: ").strip()
21            if name:
22                students.add(name)
23                print("Student added successfully.")
24            else:
25                print("Student name cannot be empty.")
26
```

Input

1
priya

Output





Q2 Menu Driven

5 marks

Write a Python menu-driven program to perform set operations.

Sample Input

Enter the first set elements: 1 2 3 4

Enter the second set elements: 3 4 5 6

Menu

1. Union

2. Intersection

3. Difference (A - B)

4. Symmetric Difference

5. Exit

Enter your choice: 1

Sample Output

Union: {1, 2, 3, 4, 5, 6}

Sample Input

Enter your choice: 2

Sample Output

Intersection: {3, 4}

Sample Input

Enter your choice: 3

Sample Output

Difference: {1, 2}

Sample Input

Enter your choice: 4



Your Code

```
1 def get_set(prompt):
2     elements = input(prompt).strip().split()
3     return set(map(int, elements))
4
5 def main():
6     print("Enter the first set elements:", end=" ")
7     A = get_set("")
8     print("Enter the second set elements:", end=" ")
9     B = get_set("")
10
11     while True:
12         print("\nMenu")
13         print("1. Union")
14         print("2. Intersection")
15         print("3. Difference (A - B)")
16         print("4. Symmetric Difference")
17         print("5. Exit")
18
19         choice = input("\nEnter your choice: ").strip()
20
21         if choice == "1":
22             print("Union:", A.union(B))
23         elif choice == "2":
24             print("Intersection:", A.intersection(B))
25         elif choice == "3":
26             print("Difference:", A.difference(B))
27         elif choice == "4":
28             print("Symmetric Difference:", A.symmetric_difference(B))
29         elif choice == "5":
30             print("Program terminated.")
```

Input

1

Output





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Q3 Exactly One Set

5 marks

Write a Python program to find the elements that appear in exactly one of three sets.

Input:

A={1,2,3}

B={3,4,5}

C={5,6,7}

Output:

{1,2,4,6,7}

Your Code

```
1 A = {1, 2, 3}
2 B = {3, 4, 5}
3 C = {5, 6, 7}
4
5 result = set()
6
7 for i in A | B | C:
8     if (i in A) + (i in B) + (i in C) == 1:
9         result.add(i)
10
11 print(result)
```

Input

stdin input

Output

{1, 2, 4, 6, 7}



← Back

Q4 Unique Words

5 marks

Write a Python program to display the unique words in a sentence using a set.

Input:

python is easy python

{'python','is','easy'}

Your Code

```
1 s = "python is easy python"
2
3 words = set(s.split())
4
5 print(words)
```

Input

stdin input

Output

```
{'python', 'easy', 'is'}
```



Q5 Unique Characters

5

Write a Python program to count the number of unique characters in a string.

Input:

Programming

Output:

8

Your Code

```
1 s = "Programming"
2
3 print(len(set(s)))
```

Input

stdin input

Output

8

Visible Test Cases

Test Case 1



Q6 Pair Sum

Write a Python program to find all pairs of elements whose sum equals a given number.

Input:

List:2 4 6 8 10

Target=12

ouput:

(4,8),(2,10)

Your Code

```
1 lst = [2, 4, 6, 8, 10]
2 target = 12
3
4 for i in range(len(lst)):
5     for j in range(i + 1, len(lst)):
6         if lst[i] + lst[j] == target:
7             print((lst[i], lst[j]))
```

Input

stdin input

Output

```
(2, 10)
(4, 8)
```



Q7 Duplicate Tuples

5 marks

Write a Python program to remove duplicate tuples from a list using a set.

Input:

[(1,2),(1,2),(3,4)]

Output:

[(1,2),(3,4)]

Your Code

```
1 def remove_duplicate_tuples(tuples_list):  
2     return list(set(tuples_list))  
3  
4 # Example  
5 inp = [(1, 2), (1, 2), (3, 4)]  
6 print(remove_duplicate_tuples(inp))
```

Input

stdin input

Output

[(1, 2), (3, 4)]



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Q8 Compare Lists

5 marks

Write a Python program to check whether two lists contain the same unique elements.

Input:

List1:1 2 2 3

List2:3 2 1

Output:

True

Your Code

```
1 list1 = list(map(int, input("List1:").split()))
2 list2 = list(map(int, input("List2:").split()))
3
4 unique1 = set(list1)
5 unique2 = set(list2)
6
7 print(unique1 == unique2)
```

Input

```
1 2 2 3
3 2 1
```

Output

```
List1:List2:True
```

Q9 Duplicate Elements

Write a Python program to find all duplicate elements in a list using a set.

Input:

10 20 30 20 10

Output:

{10,20}

Your Code

```
1 lst = [10, 20, 30, 20, 10]
2
3 seen = set()
4 duplicate = set()
5
6 for i in lst:
7     if i in seen:
8         duplicate.add(i)
9     else:
10        seen.add(i)
11
12 print(duplicate)
```

Input

stdin input

Output

{10, 20}



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Q10 First Repeated

5 marks

Write a Python program to find the first repeated element in a list using a set.

Input:

10 20 30 20 40

Output:

20

Your Code

```
1 lst = list(map(int, input().split()))
2
3 seen = set()
4 for x in lst:
5     if x in seen:
6         print(x)
7         break
8     seen.add(x)
```

Input

10 20 30 20 40

Output

20